AI Mini Project

FOREST FIRE PREDICTION

Team Number - 46

1. **Bless Singh Kukreja -** execution lead
2. **Nanabolu Bhavishya -** ppt lead
3. **Keshav Mishra -** video lead
4. **Saksham pathak -** documentation lead

Git Hub Repository Link

PROBLEM AND DATA

1 . Problem

Forest fires pose a serious threat to forests, wildlife and the environment. Early prediction can help identify potential fire risks and support timely preventive action.

2. Why is it important?

- **Helps identify potential fire risks.**
- **Supports early warning and prevention.**
- **Helps reduce damage to forests and wildlife.**

3 . Dataset

Dataset source - UCI Forest Fires Dataset(via Kaggle)

Total rows - 517

Total columns - 13

Input features - Temperature , Relative Humidity , Wind , ISI

Target variable - log_area - Log-transformed burned forest area.

OUR APPROACH

1 . DATA INPUT

The model uses temperature, relative humidity,ISI and wind speed as input features to predict the log-transformed burned forest area.

2. DATA PREPROCESSING

1. Cleaning and organizing the dataset
2. Preparing input features (temp, RH, wind,ISI)
3. Splitting data into training and testing sets

3. MODEL TRAINING

Algorithm Used: Linear Regression

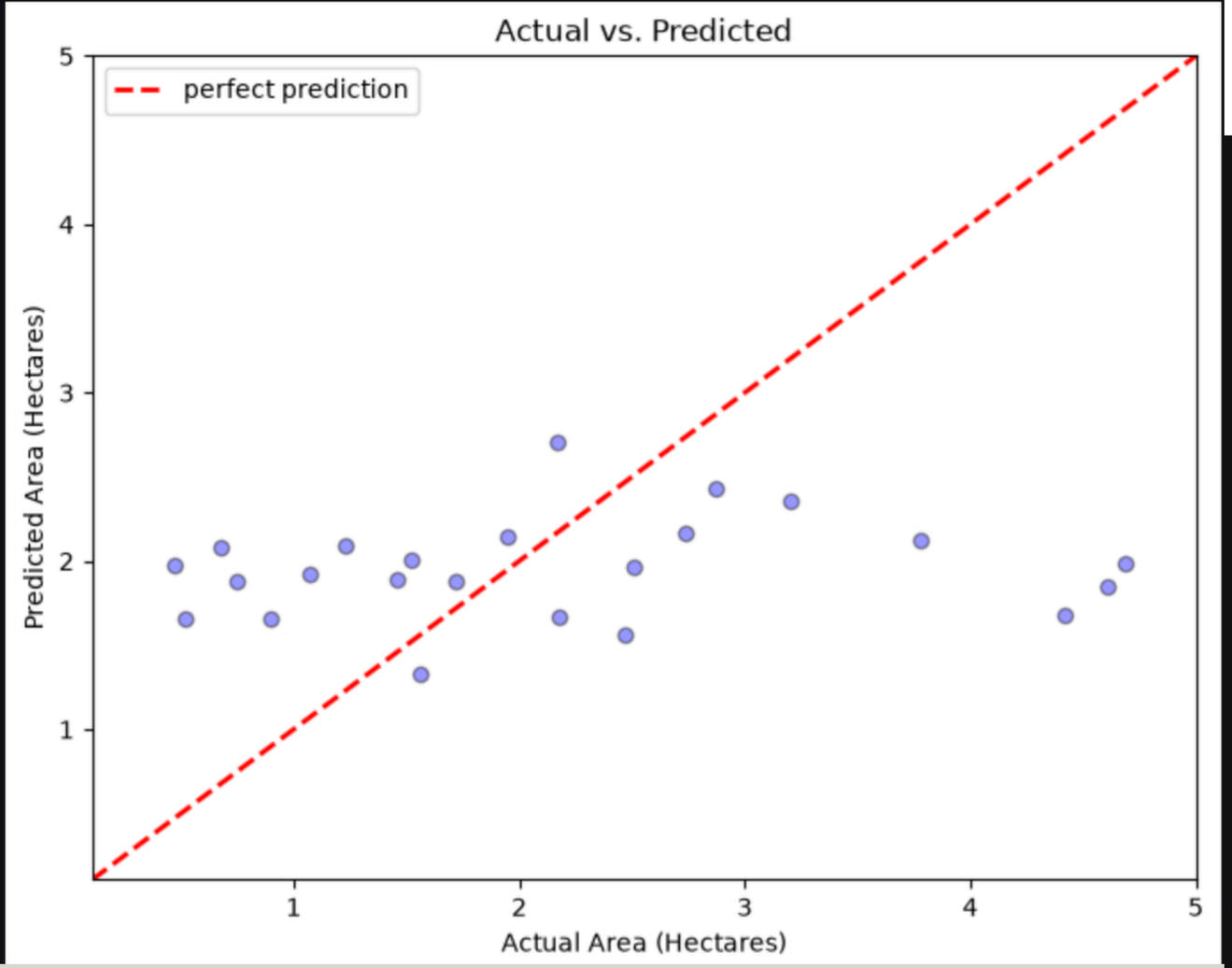
The model learns the relationships between input features and log_area

4 . PREDICTION

1. Model predicts the burned forest area.
2. Generates predicted log_area values.

Libraries Used

NumPy - Pandas - Matplotlib - Pathlib - Scikit-learn



Demo Screenshot

```
X shape: (517, 4)
y shape: (517,)
```

```python
x=relevent_df[['ISI','wind','RH','temp']]
y=relevent_df['log_area']

X_train, X_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)

model=LinearRegression()

model.fit(X_train,y_train)

y_pred=model.predict(X_test)

def compare(test,pred):
    comparison = pd.DataFrame({
                        "Actual Area": test.to_numpy(),
                        "Predicted Area": pred,
                        "Error": (test - pred)
                                                                    })
    return comparison

y_pred_actual = np.expm1(y_pred)
y_test_actual = np.expm1(y_test)

print(compare(y_test_actual,y_pred_actual).head(5))
```

[26] Python

```
     Actual Area  Predicted Area     Error
304         0.00        2.014574 -2.014574
501         0.00        2.042198 -2.042198
441         1.23        2.091167 -0.861167
153         1.46        1.888443 -0.428443
503         1.95        2.147292 -0.197292
```

Key Values

- **MAE :** 1.016
- **RMSE :** 1.277
- **MSE :** 1.629
- **R^2 SCORE :** 0.029

OUR TEAMWORK

- **BLESS SINGH KUKREJA**
  -Execution Lead
  -Data preprocessing
- **KESHAV MISHRA**
  -Video Lead
  -Model Training
- **NANABOLU BHAVISHYA**
  -PPT Lead
  -Graphing
- **SAKSHAM PATHAK**
  -Documentation Lead
  -Model Evaluation

1. Key Learnings

- Understood how machine learning can be used for forest fire data analysis.
- Learned data preprocessing, model training and prediction.
- Gained practical experience with Linear Regression.

2.What Worked

- Successfully implemented a Linear Regression model.
- Data preprocessing and model training were completed.
- Generated predictions and visualized actual vs. predicted values.

3.What Did Not Work

- Predictions showed some deviation from actual values.
- Achieving accurate predictions for all burned-area values was challenging.

4.AI Tools Disclosure

- Claude: used in first draft of README; edited and checked against the repo by Saksham..
- Gemini: Figured out how to use the highly skewed data by applying log(1+x) transform.
- chatgpt : used for organising project content.
- No AI was used to generate the dataset or the results